Year 12 Maths - Introduction to kinematics

Similar questions - Using calculus for motion in a straight line and in a plane

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle of mass 4 kg moves in a horizontal plane under a single force. At time t seconds its acceleration is $\mathbf{a} = \big((t-3)\mathbf{i} + 4\mathbf{j}\big)\ \mathrm{m\,s^{-2}}$, where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors east and north. The particle is at rest when $t = 0$.

(a) Find the values of t for which the magnitude of the force is 20 N. **[4]**

(b) Find the velocity of the particle when $t = 6$, and its speed at that time. **[4]**

2 A particle P moves on a straight line. Its displacement from a fixed point O at time t seconds is $s = 2t^3 - 15t^2 + 36t$ metres, for $t \geqslant 0$.

(a) Find expressions for the velocity and acceleration of P. **[3]**

(b) Find the times at which P is instantaneously at rest. **[2]**

(c) Find the acceleration of P at each of these times, and describe the motion of P in the interval $2 < t < 3$. **[3]**

(d) Find the total distance travelled by P in the first 4 seconds. **[4]**

Question 1

(a) $F = ma$, so $|\mathbf{a}| = \dfrac{20}{4} = 5$. $(t-3)^2 + 16 = 25$, so $t - 3 = \pm 3$ and $t = 0$ or $t = 6$

(b) $\mathbf{v} = \int \mathbf{a}\,\mathrm{d}t = \left(\frac{t^2}{2} - 3t\right)\mathbf{i} + 4t\,\mathbf{j}$ (using $\mathbf{v} = \mathbf{0}$ at $t = 0$). At $t = 6$: $\mathbf{v} = (18-18)\mathbf{i} + 24\mathbf{j} = 24\mathbf{j}$ m s^{-1}, so the speed is 24 m s^{-1} (due north)

Question 2

(a) $v = \dfrac{\mathrm{d}s}{\mathrm{d}t} = 6t^2 - 30t + 36$. $a = \dfrac{\mathrm{d}v}{\mathrm{d}t} = 12t - 30$

(b) $6(t-2)(t-3) = 0$, so $t = 2$ and $t = 3$

(c) At $t = 2$: $a = -6$ m s^{-2}; at $t = 3$: $a = 6$ m s^{-2}. For $2 < t < 3$, $v < 0$ so P moves back towards O (in the negative direction), slowing then speeding up

(d) $s(2) = 28$, $s(3) = 27$, $s(4) = 32$. P moves 28 m, then 1 m back, then 5 m forward: distance $= 28 + 1 + 5 = 34$ m